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B.Tech. Programs

Bachelor of Technology (B.Tech.)

The main objective of the B.Tech. program at IIIT-Delhi is to develop students such that they are well prepared with the necessary core competencies, problem solving and innovation skills needed to succeed in engineering/entrepreneurship careers, and are well prepared to undertake higher studies and research careers. To give due importance to both theoretical and applied aspects, the curriculum for the B.Tech. program covers the foundational aspects of the discipline, and also develops in students the engineering skills for problem solving. The B.Tech. program is broadly divided in two halves. The first half focuses on building the foundations, and is highly structured. The second half is for developing advanced/specialized skills and knowledge in various sub-areas and application domains. The program provides a lot of flexibility in selecting courses according to the student's liking and strength. For each program, there are some program specific outcomes, which are mentioned later. Besides those, there are some general program outcomes that are expected from each program. Each of the programs is expected to develop these in the students: Most engineering programs start with general courses in sciences, and then migrate to specialized courses for the disciplines. Keeping with the contemporary thinking of starting engineering courses early, the B.Tech. programs at IIIT-Delhi starts with courses in Software, Hardware, and Mathematics from first year itself. This structure empowers

students with necessary knowledge and skills early in the program. This allows students to discover the novel applications and possibilities of using their knowledge to other domains as well as learn problem solving. The courses taught in the first and second semester are same for all the programs with some program specific subjects in the second semesters. Most of the courses offered in the B.Tech. program are 4 credits each. Normally a 4-credit course requires an average effort of about 10 hours per week (including lectures). A student with full load of 5 courses in a semester should expect to put about 50 hours of effort per week during the semester. Students are required to do 156 credits to earn a B.Tech. degree. All B.Tech. programs have a defined set of core courses and many electives. Precise requirements for each B.Tech. program are given in the program specific regulations available [here](#). The B.Tech. program has an Honors option, requirements for which are as below: A B.Tech. student can also do a minor in another discipline/area. Requirements for each Minor are specified separately. Broadly, a minor degree requires the student to do about 4 to 6 courses in the minor area, taking electives or free-electives, and by doing extra credits. Currently, Minor programs are offered in Economics, Entrepreneurship, and Computational Biology.

Computer Science and Economics (CSEcon) Program

The increasing presence of digital technologies is transforming business, society, and the economy. The intersection of Economics and Computer Science is rapidly expanding, covering areas such as market design, networks, behavioral economics, financial markets, and macroeconomic modeling. Training in both disciplines is essential to grasp theoretical advancements and their industry and societal applications. The B.Tech. in Computer Science and Economics at IIIT-Delhi offers rigorous training at the intersection of economics and computation. It prepares students for careers in banking, finance, consulting, tech, and IT services while also providing a strong foundation for graduate studies. With core courses in Computer Science,

Economics, and Mathematics, along with diverse electives, the program blends quantitative and computational skills for real-world applications. IIITD leverages its expertise in Computer Science and interdisciplinary education to offer a forward-thinking program which is also aligned with the National Education Policy, 2020. At the end of this program, a student should have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The program follows the philosophy of having a small set of core courses and many electives allowing students significant flexibility in designing their curriculum and specialization. The overall program structure is given in the table below:

Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6 onwards
Introduction to Programming	Data Structures and Algorithms	Operating Systems	Algorithm Design and Analysis	Game Theory [Special Elective - Bucket I]	Digital Circuits
Principles of Economics	Macroeconomics	Convex Optimization	Econometrics II [Special Elective - Bucket II]	Maths-I (Linear Algebra)	Maths-II (Probability and Statistics)
Discrete Mathematics	Database Management Systems	Introduction to HCI	Computer Organization	Advanced Programming	Econometrics I
Technical Communication+	Environmental Sciences	Communication Skills	SSH Elective	Math-III (Multivariate Calculus)	Microeconomics

Note: Special Electives mentioned in [] can be done any semester. However, the prerequisites for the courses must be met first. Students must do at least one course from each special elective bucket, defined as follows:

Bucket 1	Bucket 2
Game Theory	Maths-I (Linear Algebra)
Macroeconomics	Maths-II (Probability and Statistics)
Convex Optimization	Discrete Mathematics
Econometrics II [Special Elective - Bucket II]	Database Management Systems
Math-III (Multivariate Calculus)	Introduction to HCI
Microeconomics	Computer Organization
	Advanced Programming

The list of courses in buckets I and II may be updated as needed. For details please see Program Regulations

Admission through JEE Mains. Fees as per Institute norms for the B.Tech Programs.

Computer Science and Engineering (CSE) Program

Computer Science is a powerful engineering tool for problem solving in a variety of domains. IIIT-Delhi has developed a CSE program, which, besides developing the

computing sciences foundations, also develops in students the engineering skills for problem solving using computing sciences. The program at IIIT-Delhi - "inverts the pyramid" - and start with computing-oriented courses first, and then provides flexibility for taking a variety of courses later. This empowers the students to explore innovative applications of computing in the initial years of the program so they can apply computing techniques in different domains. Program Objectives for the CSE program are to help develop the following attributes in students (in addition to the general attributes mentioned above): The set of core courses offered in B.Tech. CSE program are shown in the table below (courses mentioned in [] are electives and actual courses for these slots are as defined from semester to semester.) For students of 2019 and previous batches, program structure is available [here](#). In the 3rd and 4th year, most of the courses are electives. Some of these electives have to be CSE electives, some have to be Social Sciences and Humanities electives, and the rest are Open Electives (i.e. any course can be taken). Detailed regulations about the program, including the requirements for graduation and the Honors program, are available [here](#).

Electronics and Communications Engineering (ECE) Program

As a discipline, ECE focuses on the design of underlying hardware systems. Our curriculum is directed to applications in major areas such as telecommunications, energy and electronics sectors, while encouraging development of necessary skills for integration of hardware and software components. We believe that many creative opportunities exist at the boundaries of traditional CSE and ECE, and have accordingly planned for cross-training of students across disciplinary boundaries. The curriculum for ECE therefore has many courses in common with the CSE program initially. The set of core courses are shown in the table below (courses mentioned in [] are electives and actual courses for these slots are as defined from semester to semester.) For students of 2019 and previous batches, program structure is given [here](#). Most courses in Sem 5th

to 8th are electives (an elective course is one which is not compulsory, and a student may have choices from which to select the courses he/she wants to do.). 3 or more elective courses may be chosen from various areas like Circuits and VLSI, Communication Engineering, Signal & Image Processing and Control & Embedded Systems etc., which allow a student to focus on some areas and gain a deeper knowledge and skills in those. Detailed regulations about the program, including the requirements for graduation and the Honors program, are available [here](#).

Computer Science and Applied Mathematics (CSAM) Program

The increasing use of sophisticated mathematical tools and techniques in tandem with computational tools in several areas such as computational finance, biology, e-commerce, weather forecasting, and data science motivates the need for a program that will produce graduates with computational skills as well as the ability to use sophisticated mathematical concepts and tools in order to tackle these problems. The Computer Science and Applied Mathematics program aims to develop such graduates. The program is similar to the Mathematics and Computing programs operating in many leading Institutions. The program has a small set of core courses in both Computer Science and Mathematics, and many electives which can be taken from both the disciplines. This enables the students to build a program most suitable for them.

Program Objectives: At the end of this program, a student should have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The B.Tech. program in CSAM follows the philosophy of having a small set of core courses and many electives allowing students significant flexibility in designing their curriculum and specialization. The overall program structure is given in the table below: For students of 2019 and previous batches, program structure is given [here](#). In the final year, all courses are electives. Details about the structure of the program and the requirements for graduation are given [here](#).

Computer Science and Design (CSD) Program

The B.Tech. in Computer Science (CS) and Design aims to develop graduates that are not only well versed with computing approaches, tools, and technologies, but are also experienced with Design approaches and new Media technologies and uses. The program has a small set of core courses in CS and Design, and many electives which can be taken from CS as well as Design and Digital Media. This enables the students to build a program most suitable for them. The program will prepare students to work in the IT industry as well as digital media industry like gaming, animation, virtual/augmented reality, etc. The program will also allow students, who want to pursue higher studies, to take up higher studies in CS/IT or in Design. The program aims to develop capabilities in CS as well as Design and Digital Media. At the end of this program, a student should have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The overall program structure is given in the table below – * Students who will do Discrete Mathematics in semester 3 will be allowed to do ADA. Also ADA and ADA(B) are anti-requisites. For students of 2019 to previous batches, program structure is given here. In the 6th, 7th and 8th Semester all courses are electives. Details about the structure of the program and the requirements for graduation are given here.

Computer Science and Social Sciences (CSSS) Program

This unique program, B.Tech. in Computer Science (CS) and Social Sciences (SS), aims to develop computer science engineers with strong understanding of relevant social science disciplines. The program will also allow a student to pursue further studies in social sciences, besides allowing them to pursue higher studies in CS/IT, as well as many interdisciplinary programs. As it is a 4 year program, it will satisfy the requirements of almost all higher studies programs in India as well as overseas. It may be an ideal program for those students who are not sure if they want to pursue

engineering careers and would like to explore the possibility of going for social sciences later, but want to be ready to take computer science career if desired. The program aims to develop capabilities in Computer Science as well as Social Science. At the end of this program, a student should have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The B.Tech. program at IIITD follows a philosophy of having a small set of core-courses, allowing students significant flexibility in designing their curriculum and specialization. In the first few semesters mostly core courses are done. The structure for first few semesters is as below, courses mentioned in [] are electives and actual courses for these slots are as defined from semester to semester: For students of 2019 and previous batches. [Clickhere](#) In the 6th, 7th and 8th Semester all courses are electives. Details about the structure of the program and the requirements for graduation are [givenhere](#).

Computer Science and Biosciences (CSB) Program

With the advent of high-throughput techniques, biological sciences are grappling with a paradigm shift towards data-intensive explorations and challenges for management and analysis of massive data. Apart from fundamental contributions to basic science, data driven analysis in biology has the potential to conquer challenges such as modeling and control of complex diseases, management and diagnosis of pathologies, personalized medicine, drug and vaccine design, among others. Making progress on these frontiers requires insight into biological processes, algorithms, machine learning techniques, mathematical modeling, apart from numerical and programming skills. Thus, interdisciplinary education that imparts knowledge of foundations of biology and computer science as well as training in modeling and analysis of biomedical data is the key to create personnel who can provide solutions to problems on the interface of computation and biology. Knowledge of different aspects of modern biology and computational sciences will facilitate addressing relevant problems in biology and

medicine. Towards this aim, an undergraduate program that seamlessly integrates foundations of computer science, biology and mathematics along with training to ask data-driven questions in biology and medicine is an important step in this direction. The program aims to develop capabilities in Computer Sciences as well as in Biosciences. At the end of this program, a student should have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The B.Tech. program at IIITD follows a philosophy of having a small set of core-courses, allowing students significant flexibility in designing their curriculum and specialization. Majority of core courses are completed in the first four semesters. The structure for first few semesters is as follows: For students of 2019 and previous batches. [Clickhere](#) In the 6th, 7th and 8th Semester all courses are electives. Details about the structure of the program and the requirements for graduation are [givenhere](#).

Computer Science and Artificial Intelligence (CSAI) Program

Artificial Intelligence (AI) has become an integral part of technology in our daily lives, driving to office, searching for a restaurant, getting news updates, and recommendations on social media are all using AI. With increase in usage, there is a significant requirement of researchers who can understand AI and build AI technologies. This program will provide students an opportunity to learn both foundational and experimental components of AI and Machine Learning. A student completing this program will be able to undertake industry careers involving innovation and problem solving using Artificial Intelligence (AI) and Machine Learning (ML) technologies and research careers in AI, ML, and, in general, Computer Science areas. Along with courses that provide specialization in AI, students will also have an option to explore applied domains such as computer vision, natural language processing, robotics, and autonomous systems as well as other interdisciplinary areas such as neuroscience, edge computing, and Internet of Things. At the end of this program, a student should

have following attributes (in addition to the general attributes mentioned on B.Tech. Page): The Foundation program provides the basic knowledge about Computer Science and Artificial Intelligence (CSAI) through a set of core courses, which are compulsory for all students. The set of core courses are shown in the table below (courses mentioned in [] are electives and actual courses for these slots are as defined from semester to semester.) For students of 2019 and previous batches. [Clickhere](#) For B.Tech. (CSAI), the program structure & requirements for graduation and the Honors program, are givenhere.

Electronics and VLSI Engineering (EVE) Program

IIIT Delhi aims to encourage research and innovation in Information Technology (IT) and allied areas. The objective of the B.Tech. program in Electronics and VLSI Engineering (EVE) is to prepare students to undertake careers involving innovation and problem solving using suitable techniques and hardware and software technologies, or to undertake advanced studies for research careers. In order to give due importance to applied as well as theoretical aspects of EVE, the curriculum for the B.Tech. (EVE) program covers most of the foundational aspects and also develops in students the engineering skills for problem solving. Towards this, the B.Tech. (EVE) program at IIIT-Delhi starts with computing and Electronics courses first, and allows the possibility of doing science courses later. Besides being better suited for developing engineering capabilities, it also enables the possibility of students seeing newer applications and possibilities of using computing and electronics in these subjects. TBA The first semester of the EVE program is common with all - this allows flexibility to students in moving from one program to the other. The second and third semester is common with the ECE program. The fourth and fifth semester of the program is relatively fixed, comprising mostly core courses for the program. From sixth semester onward the program can be mostly flexible, consisting of electives. Last updated: 28-03-2025